Method 2:
Let $y = 2x - 3$. Then $x = \dfrac{y+3}{2}$. Let $g(y) = \dfrac{y+3}{2}$.

Now

$$(g \circ f)(x) = g(f(x)) = g(2x-3) = \frac{(2x-3)+3}{2} = x.$$

$$(f \circ g)(y) = f(g(y)) = f\left(\frac{y+3}{2}\right) = 2\left(\frac{y+3}{2}\right) - 3 = y.$$

Thus, $g \circ f = I_X$ and$f \circ g = I_Y$

This implies that f and g are bijections and inverses to each other. Hence f is a bijection and $f^{-1}(y) = \dfrac{y+3}{2}$. Replacing y by x we get, $f^{-1}(x) = \dfrac{x+3}{2}$.

1.6.6 Algebra of Functions

A function whose co-domain is $\mathbb{R}$ or a subset of $\mathbb{R}$ is called a real valued function. We can discuss many more operations on functions if it is real valued.

Let f and g be two real valued functions. Can we define addition of f and g? Naturally we expect the sum of two functions to be a function. The value of $f + g$ at a point x should be related to the values of f and g at x. So to define $f + g$ at a point x, we must know both $f(x)$ and $g(x)$. In other words x must be in the domain of f as well as in the domain of g. And the natural way of defining $f + g$ at x is $f(x) + g(x)$. So if we impose a condition that the domains of f and g to be the same, then we can define $f + g$. In the same way we can define subtraction, multiplication and many more algebraic operations available on the set $\mathbb{R}$ of the real numbers.

Definition 1.11

Let X be any set. Let f and g be real valued functions defined on X. Define, for all $x \in X$

- $(f + g)(x) = f(x) + g(x)$.
- $(f - g)(x) = f(x) - g(x)$.
- $(fg)(x) = f(x)g(x)$.
- $\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}$, where $g(x) \neq 0$.
- $(cf)(x) = cf(x)$, where c is a real constant.
- $(-f)(x) = -f(x)$.

Note that the domain may be any set, not necessarily a set of numbers. For example if X is a set of students of a class, f and g functions representing the marks obtained by the students in two tests, then the function $f + g$ represent the total marks of the students in the two tests. It is easy to see that the operations addition, subtraction, multiplication and division defined above satisfy the following properties.

- $(f + g) + h = f + (g + h)$
- $f + g = g + f$
- $\mathbf{0} + f = f + \mathbf{0}$, where $\mathbf{0}$ is the zero function defined by $\mathbf{0}(x) = 0$ for all x.
- $f + (-f) = (-f) + f = \mathbf{0}$
- $f(g + h) = fg + fh$
- $(c_1 + c_2)f = c_1 f + c_2 f$ where c_1 and c_2 are real constants.

We can list many more properties of these operations. The proofs are simple; however let us prove only one to show a way in which these properties can be proved.

Let us prove $f(g + h) = fg + fh$. To prove $f(g + h) = fg + fh$ we have to prove that $(f(g + h))(x) = (fg + fh)(x)$ for all x in the domain.